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Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: Analytical study

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ABSTRACT

The undesirable vortex-induced vibration (VIV) may seriously influence the fatigue life and serviceability of bridge structures. It is important to take countermeasures to suppress the adverse VIV of long-span bridges. In the present study, a novel inerter-based system, namely the tuned mass damper inerter (TMDI), is proposed to control the VIV of the main deck of long-span bridges. In this system, an inerter device, which is able to transform the linear motion into the high-speed rotational motion and thus significantly amplifies the physical mass of the system, is incorporated into the conventional tuned mass damper (TMD) system to further improve the performance of TMD. An applicable layout of the TMDI inside the bridge deck is introduced and the governing equations of the structure-TMDI system subjected to VIV are established. The optimization of the TMDI parameters with the consideration of nonlinear aeroelastic effect is derived. The control performance and robustness of the proposed system are investigated through an analytical case study in both the time and frequency domains. It is observed that the TMDI system can obviously reduce the VIV responses of the bridge deck. Moreover, compared to the conventional TMD system, the static stretching of the spring due to gravity and the oscillation amplitude of the mass block in the TMDI system are significantly reduced. These properties make the proposed TMDI system an attractive alternative for the VIV control of long-span bridges.

1. Introduction

With the ever growth of span length, bridge structures are becoming more and more flexible and therefore more and more susceptible to the wind effects. Vortex-induced vibration (VIV), a typical wind-induced vibration, may seriously influence the fatigue life and serviceability of bridge structures [1–5]. In certain circumstances, the excessive VIV may cause catastrophic damages to the important components of the bridge structure or even collapse of the bridge. It is therefore by all means necessary to take countermeasures to suppress the adverse VIV responses of bridge structures.

Extensive research works have been conducted on the VIV control of bridges, and a lot of countermeasures have been proposed. These methods can be roughly classified into two categories [6]: the aerodynamic countermeasures and the mechanical methods. Rarely, a third method through changing the structural properties such as increasing the stiffness in the desired direction or adding intermediate supports may also be considered as a possible option [7].

The aerodynamic countermeasure to attenuate the VIV responses of a bridge lies in retrofitting the cross-section of the main deck. A bridge deck with a favorable aerodynamic configuration will significantly reduce the vortex-shedding effect and thereby reduce the VIV responses of the bridge. The aerodynamic configuration of the main deck is usually achieved by attaching additional components on the deck, such as the wind fairings [8], baffle plates [9] and guide vanes [4,10], or through adjusting the configuration and/or location of some non-structural components, such as crash barriers, railings, and maintenance rails [6,11,12]. However, it should be noted that certain limitations exist in the aerodynamic countermeasures. For example, bridge structure is normally regarded as the landmark for a city, the installation of the additional aerodynamic components will influence the aesthetics of the structure, and the installation and maintenance of these additional components can be very expensive. Moreover, due to the complexity of deck shapes, there is no general countermeasure that is applicable for all deck shapes. A countermeasure that is suitable for one bridge deck may show adverse effect for another [13]. Furthermore, the effectiveness of some countermeasures depends not only upon the configuration of the primary structure but also the local wind condition. For example, Ge et al. [11] concluded that the control effects of guide vanes are closely related with the wind angle of attack.

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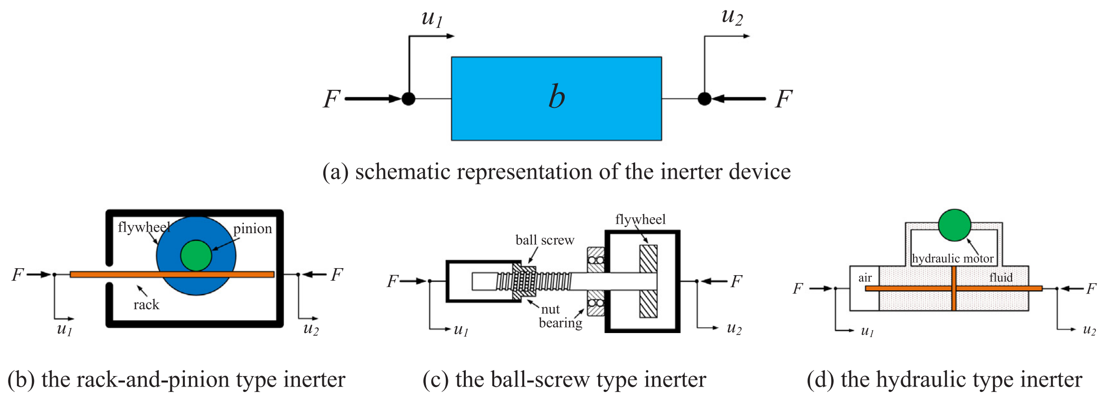


Fig. 1. (a) Schematic representation of the two-terminal inerter device and (b, c and d) some typical inerter devices.

Among the mechanical countermeasures, the tuned mass damper (TMD) is most widely adopted (e.g. [1,14–17]) due to its simplicity and effectiveness. A TMD is a device consisting of a mass, a spring and a dashpot that is attached to the vibrating primary structure to attenuate the undesirable vibrations [18,19]. The natural frequency of the TMD is tuned to the fundamental vibration frequency of the primary structure so that the damper will resonant out of phase with the original structure and a large amount of the structural vibrating energy is transferred to the TMD and then dissipated by the damper. However, the application of TMDs in the bridge structures also faces some practical challenges. The first challenge is the large static stretching of the spring due to gravity. For the hanging-type TMD, the static stretching of the spring can be estimated as $x_0 = g/\omega_{TMD}^2$ [20], where g is the acceleration of gravity and ω_{TMD} is the circular frequency of the TMD, which is close to the vibration frequency of the primary structure to be controlled as discussed. The static stretching of the spring can be quite large for a bridge with low natural frequencies. The second challenge arises from the energy dissipation principal of TMD, i.e. the vibration energy of the main structure is transferred to the TMD as mentioned above, which makes the stroke of the TMD not small. These characteristics of TMD make it difficult to be applied in engineering practice when the TMDs are to be installed inside a box girder, especially when the streamlined deck shapes are adopted and the height of the bridge deck is limited.

Recently, a novel device dubbed inerter was developed and used to control the vibrations of engineering structures. This two-terminal device, originally proposed by Smith in the early 2000s [21], is able to transform the linear motion into the high-speed rotational motion and thus significantly amplifies the physical mass of the system. It is therefore can generate an inertial force that is two orders of magnitude higher than that generated by its physical mass [22]. This device was initially employed in the suspension system of high-performance vehicles such as Formula-1 racing cars [23]. The inerter-based device was then introduced to control the vibrations of civil engineering structures by Wang et al. [24,25]. Ikago et al. [26] proposed a tuned viscous mass damper (TVMD) system to mitigate the seismic induced vibrations of buildings. The closed-form solution of an optimum design for the TVMD was derived when it is installed in a single-degree-of-freedom (SDOF) structure. Lazar et al. [27] incorporated the inerter device with the damping and stiffness elements to form a tuned inerter damper (TID), and installed it between the adjacent storeys of a building for seismic-induced vibration control. Their results indicated that a high level of vibration control can be achieved with a low amount of added mass, which makes the TID an attractive potential alternative to the traditional TMDs. Marian and Giaralis [28] proposed to combine the inerter device with the conventional tuned mass damper (TMD) to form a tuned mass damper inerter (TMDI) system to reduce the vibrations of stochastically support-excited structural systems. Their results indicated that, with the same amount of attached mass, the TMDI system is more effective than the TMD system in reducing the base-excited vibration of the primary structure. Wen et al. [29] and Pietrosanti et al. [30] further compared and discussed the performances of different inerter-based dampers with different design methodologies (i.e. TVMD, TMDI and TID) in reducing the seismic-induced vibrations of structures. Besides the applications of inerter-based dampers in the control of seismic-induced vibrations, this kind of device has also been applied to the vibration control of cables [31–33], wind-induced vibration of tall buildings [34], and wave-induced heave motion of offshore semi-submersible platforms [35]. To the best knowledge of the authors, the application of the inerter-based system for VIV control of long-span bridges has never been reported.

This paper proposes using TMDI to control the VIV of long-span bridges. An applicable layout of the TMDI inside the bridge deck is introduced and the optimal TMDI parameters are derived. It should be noted that VIV is a nonlinear self-excited aeroelastic phenomenon, the influence of the nonlinear aeroelastic effect is considered in the optimization of the TMDI parameters. The control performance and robustness of the proposed system are investigated through an analytical case study in both the time and frequency domains. The arrangement of this paper is as follows: Section 2 introduces the layout of the TMDI installed inside the bridge deck, and the governing equations of the structure-TMDI system subjected to the VIV is established; The optimization of the TMDI system with the consideration of the nonlinear aeroelastic effect is derived in Section 3; Sections 4 and 5 report the performances of this device in the bridge VIV control; and Section 6 summarizes the major findings of the present work.

2. Analytical model

2.1. Layout of the TMDI system

Inerter is a two-terminal device that can generate an apparent mass that is much larger than its physical mass through transforming the linear motion into the high-speed rotational motion of the system. Up to now, several types of inerter have been proposed by different researchers. For example, the rack-and-pinion type [36], the ball-screw type [37] and the hydraulic type inerter [38] were widely used in the engineering practices. To facilitate the interpretation of this device, Fig. 1 depicts its schematic representation and the operating principles for the above mentioned three typical types of inerter. It can be seen that this device has two terminals, and both ends of the device should be connected to the structure or the ground. For an ideal linear inerter, the force generated by the device can be calculated as [21]:

$$F = b(\ddot{u}_1 - \ddot{u}_2) \tag{1}$$

in which b is the inertance (apparent mass), which attains the unit of mass that can be two orders of magnitude higher than its physical mass [22]; $\ddot{u}_1 - \ddot{u}_2$ is the relative acceleration between the inerter's two nodes.

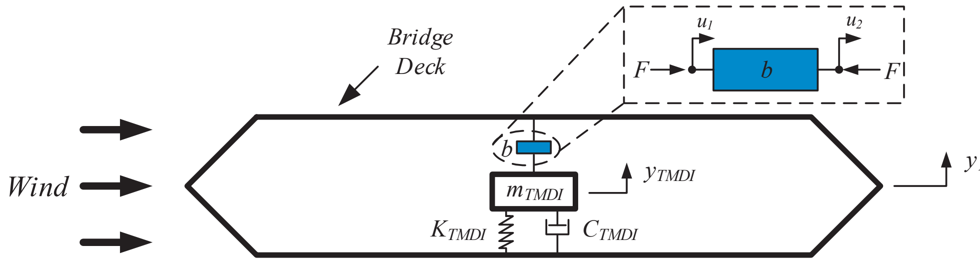


Fig. 2. Single-degree-of-freedom primary structure incorporating with a TMDI system.

To best use the mass amplification effect of the device, one end of the device is connected to the structure to be controlled while the other end is normally connected to the ground ([28,30]). To control the VIV of the main deck of a long-span bridge, it is difficult to connect one end of the device to the ground due to the high elevation of the bridge deck. In the present study, the TMDI system is proposed to be installed inside the box cell of the main girder-an arrangement that is commonly adopted when the conventional TMDs are used for the VIV control of the bridge girder (e.g. in [1]). For the bridge girder as shown in Fig. 2, when the wind with a velocity of U arrives at the bridge, vortex shedding will result in vibrations in the vertical direction of the bridge girder. A conventional TMD is installed within the box cell of the main girder. One end of the inerter (with an inertance of b) is connected to the TMD and the other end is connected to the box cell. It should be noted that this arrangement (i.e. with no end connecting to the ground) is also adopted by some other researchers. For example, Lazar et al. [27] and Wen et al. [29] used similar arrangements to control the earthquake-induced vibrations of tall buildings.

2.2. Governing equations of motion

With the TMDI arrangement as shown in Fig. 2, the equations of motion of the system can be written in the following form:

$$\begin{bmatrix} m_1 + m_{TMDI} & m_{TMDI} \\ m_{TMDI} & m_{TMDI} + b \end{bmatrix}\begin{bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} C & \\ & C_{TMDI} \end{bmatrix}\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} + \begin{bmatrix} K & \\ & K_{TMDI} \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} F_{wind} \\ 0 \end{bmatrix} \tag{2}$$

where m_{TMDI} is the mass of the TMDI mass block and m_1 is the mass of the primary structure; C_{TMDI} and K_{TMDI} denote the damping and stiffness of the TMDI system, respectively; C and K are the corresponding values for the bridge structure; y_1 is the displacement of the deck; $y_2 = y_{TMDI} - y_1$ denotes the relative displacement between the TMDI and the main deck; F_{wind} is the wind force acting on the deck induced by vortex shedding, which can be estimated by the following nonlinear semi-empirical model [39]:

$$F_{wind} = \frac{1}{2}\rho U^2 D\left[Y_1 \cdot \left(1 - \varepsilon\frac{y_1^2}{D^2}\right)\frac{\dot{y}_1}{U} + Y_2\frac{y_1}{D} + \tilde{C}_L(t)\right] \tag{3}$$

in which $\tilde{C}_L(t)$ denotes the lift coefficient resulting from the vortex-shedding effect; ρ is air density; U is wind velocity; D is deck height; Y_1, Y_2, and ε are aeroelastic parameters that can be obtained from experimental tests.

Substituting Eq. (3) into Eq. (2), Eq. (2) then can be rewritten as:

$$\begin{bmatrix} m_1 + m_{TMDI} & m_{TMDI} \\ m_{TMDI} & m_{TMDI} + b \end{bmatrix}\begin{bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} C - C_{ae} & \\ & C_{TMDI} \end{bmatrix}\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} + \begin{bmatrix} K - K_{ae} & \\ & K_{TMDI} \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{\rho U^2 D \tilde{C}_L(t)}{2} \\ 0 \end{bmatrix} \tag{4}$$

where $C_{ae} = \rho U Y_1(D^2 - \varepsilon y_1^2)/2D$ denotes the aeroelastic damping and $K_{ae} = \rho U^2 Y_2/2$ is the aeroelastic stiffness.

To facilitate the derivation of the optimal parameters for the TMDI system, Eq. (4) can be written in the modal coordinate as follows:

$$\begin{bmatrix} M_i + \mu M_i & \mu M_i \\ \mu M_i & \mu M_i + \beta M_i \end{bmatrix}\begin{bmatrix} \ddot{\eta}_i \\ \ddot{\eta}_{TMDI} \end{bmatrix} + \begin{bmatrix} 2M_i\omega_{ni}(\xi_{ni} - \xi_{ae}) & \\ & 2\mu M_i \omega_{TMDI}\xi_{TMDI} \end{bmatrix}\begin{bmatrix} \dot{\eta}_i \\ \dot{\eta}_{TMDI} \end{bmatrix} + \begin{bmatrix} M_i\omega_{ni}^2 - K_{ae} & \\ & \mu M_i \omega_{TMDI}^2 \end{bmatrix}\begin{bmatrix} \eta_i \\ \eta_{TMDI} \end{bmatrix} = \begin{bmatrix} F_i \\ 0 \end{bmatrix} \tag{5}$$

where $M_i = \bar{m}\int_0^L \phi_i(x)^2 dx$ is the modal mass for the ith mode with $\bar{m}$ denoting the equivalent mass of the deck per unit length, L is the total length of the bridge and $\phi_i(x)$ is the modal shape of the deck for the ith mode; η_i is the generalized coordinate of the ith mode; η_{TMDI} is the relative displacement of the TMDI; ξ_{ni} and ω_{ni} denote the modal damping ratio and modal frequency of the primary structure for the ith mode, respectively; ω_{TMDI} and ξ_{TMDI} are the corresponding values for the TMDI system; μ and β are the mass ratios of the TMDI mass block and inertance with respect to the modal mass of the bridge, which are defined as:

$$\begin{aligned} \mu &= m_{TMDI}/M_i \\ \beta &= b/M_i \end{aligned} \tag{6}$$

ξ_{ae} and K_{ae} are the aeroelastic damping coefficient and stiffness that can be calculated as:

$$\begin{aligned} \xi_{ae} &= \frac{1}{2M_i\omega_{ni}}\left[\frac{\rho U^2 D}{2}Y_1\frac{\int_0^L \phi_i^2(x)dx}{U} - \frac{\rho U^2 D}{2}Y_1\varepsilon\frac{\int_0^L \phi_i^4(x)dx}{D^2 U}\eta_i^2\right] \\ K_{ae} &= \frac{\rho U^2 D}{2}Y_2\frac{\int_0^L \phi_i^2(x)dx}{D} \end{aligned} \tag{7}$$

and F_i is the generalized lift force resulting from vortex-shedding effect, which can be determined by:

$$F_i = \left\{\int_0^L \int_0^L \left(\frac{1}{2}\rho U^2 D\right)^2 \tilde{C}_L(x_1, t)\tilde{C}_L(x_2, t)R_{cor}(\Delta x)dx_1 dx_2\right\}^{1/2} \tag{8}$$

where $R_{cor}(\Delta x) = R_{cor}(|x_1 - x_2|)$is the spatial coherence function for the vortex-shedding to consider the differences between the wind loads at locations x_1 and x_2 along the main deck.

It is noticeable that by setting $\beta = 0$ in Eq. (5), the equations of motion for the structure-TMDI system degrade to the conventional structure-TMD system. In this sense, the TMDI system can be regarded as a generalization of the conventional TMD system.

2.3. Response of the TMDI system

The modal frequency-response-functions (FRFs) for the displacement of the girder $H_{\eta_i}(f)$ and the relative displacement of the TMDI $H_{\eta_{TMDI}}(f)$ can be determined as follows based on Eq. (5):

$$\begin{bmatrix} H_{\eta_i}(f) \\ H_{\eta_{TMDI}}(f) \end{bmatrix} = [D]^{-1}\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{9}$$

where

$$[D] = -(2\pi f)^2\begin{bmatrix} M_i+\mu M_i & \mu M_i \\ \mu M_i & \mu M_i+\beta M_i \end{bmatrix} + i(2\pi f)\begin{bmatrix} 2M_i\omega_{ni}(\xi_{ni}-\xi_{ae}) & \\ & 2\mu M_i\omega_{TMDI}\xi_{TMDI} \end{bmatrix} + \begin{bmatrix} M_i\omega_{ni}^2 - K_{ae} & \\ & \mu M_i\omega_{TMDI}^2 \end{bmatrix} \tag{10}$$

and $i=\sqrt{-1}$.

The response spectra of the main deck $S_{\eta_i}(f)$ and the TMDI $S_{\eta_{TMDI}}(f)$ therefore can be obtained as:

$$\begin{bmatrix} S_{\eta_i}(f) \\ S_{\eta_{TMDI}}(f) \end{bmatrix} = \begin{bmatrix} H_{\eta_i}^*(f)\cdot H_{\eta_i}(f) \\ H_{\eta_{TMDI}}^*(f)\cdot H_{\eta_{TMDI}}(f) \end{bmatrix}\cdot S_F(f) \tag{11}$$

in which, * denotes the complex conjugation; $S_F(f)$ is the power spectral density (PSD) function of the vortex-shedding effect on the main deck. For long-span bridges, the integral length scale of the vortices is small compared to the wind exposed length of the main deck L, $S_F(f)$ thus can be expressed as follows [40]:

$$S_F(f) = \frac{2q_V^2D^3\sigma_{CL}^2\lambda}{\sqrt{\pi}f_sB}\exp\left[-\left(\frac{1-f/f_s}{B}\right)^2\right]\int_0^L\phi_i^2(x)dx \tag{12}$$

where $q_V=\rho U^2/2$ is the wind pressure; σ_{CL} is the root mean square (RMS) value of the lift coefficient $\widetilde{C}_L(t)$; $f_s=US_t/D$ is the vortex-shedding frequency with S_t denoting the Strouhal number; B is the load spectrum bandwidth; and λ is a dimensionless parameter related to the coherence function (i.e. the coherency length-scale).

The corresponding mean square values of η_i and η_{TMDI} are therefore:

$$\begin{bmatrix} \sigma_{\eta_i}^2 \\ \sigma_{\eta_{TMDI}}^2 \end{bmatrix} = \int_f\begin{bmatrix} S_{\eta_i}(f) \\ S_{\eta_{TMDI}}(f) \end{bmatrix}df \tag{13}$$

3. Optimization of the TMDI system

To achieve the best control performance of the system, the TMDI parameters should be optimized. In this study, the mean square motion of the main deck is of interest. Thus, the optimization of the TMDI system is to obtain the minimum value of the following J performance index:

$$J = \sigma_{\eta_i}^2 = \int_{-\infty}^{\infty}S_{\eta_i}(f)df \tag{14}$$

in which the response spectrum of the main deck $S_{\eta_i}(f)$ can be obtained from Eq. (11).

It should be noted that the aeroelastic damping ξ_{ae} in Eq. (7) is a function of the structural response (i.e. it is motion-dependent), which makes the optimization very complicated. To simplify the analysis, the aeroelastic damping model proposed by Vickery and Basu [41,42] is adopted in the present study, in which the aeroelastic damping is modelled as a function of the mean square motion of the deck, i.e.:

$$\xi_{ae} = K_a\frac{\rho D^2}{\bar{m}}\left[1-\left(\frac{\sigma_{\eta_i}}{Da_L}\right)^2\right] \tag{15}$$

where σ_{η_i} is the RMS of the deck response η_i; K_a is the linear aeroelastic damping parameter (with the maximum value of $K_{a\,\max}$); a_L is the nonlinear aeroelastic damping parameter that limits the "lock-in" displacements of the bridge deck. The aerodynamic parameters K_a and a_L in Eq. (15) can be obtained through the experimentally identified parameters Y_1 and ε in Eq. (7), which will be further discussed in the following section.

Since the aeroelastic damping is motion-dependent, according to Eqs. (13) and (15), it is obvious that an iteration procedure is needed in order to calculate the values of σ_{η_i} and $\sigma_{\eta_{TMDI}}$. Moreover, due to the nonlinear self-excited characteristic of the aeroelastic damping, there is a possibility that the calculated result at some design parameters may generate a negative modal damping in the structure-TMDI system. This non-realistic result will lead to the instability of the structure-TMDI system [43,44]. Thus, a constraint condition should be added in the process of iteration to avoid the non-realistic solutions. As indicated in [43,44], the constraint condition for the structure-TMDI system is that the real parts of the complex eigenvalues of this system should be negative.

The complex eigenvalues of the structure-TMDI system can be evaluated as solutions of the following characteristic equation:

$$\det\begin{bmatrix} \lambda^2(1+\mu)M_i + 2M_i\omega_{ni}(\xi_{ni}-\xi_{ae})\lambda + M_i\omega_{ni}^2 & \lambda^2\mu M_i \\ \lambda^2\mu M_i & \lambda^2(\mu+\beta)M_i + 2\mu M_i\omega_{TMDI}\xi_{TMDI}\lambda + \mu M_i\omega_{TMDI}^2 \end{bmatrix} = 0 \tag{16}$$

that is

$$\begin{aligned} &\lambda^4[(1+\mu)(\mu+\beta)M_i^2 - \mu^2M_i^2] \\ &+ \lambda^3[2M_i^2\omega_{ni}(\xi_{ni}-\xi_{ae})(\mu+\beta) + 2\mu M_i^2\omega_{TMDI}\xi_{TMDI}(1+\mu)] \\ &+ \lambda^2[M_i^2\omega_{ni}^2(\mu+\beta) + (1+\mu)\mu M_i^2\omega_{TMDI}^2 \\ &+ 4\mu M_i^2\omega_{ni}(\xi_{ni}-\xi_{ae})\omega_{TMDI}\xi_{TMDI}] \\ &+ \lambda[2\mu M_i^2\omega_{TMDI}\xi_{TMDI}\omega_{ni}^2 + 2M_i\omega_{ni}(\xi_{ni}-\xi_{ae})\mu M_i\omega_{TMDI}^2] \\ &+ \mu M_i^2\omega_{ni}^2\omega_{TMDI}^2 = 0 \end{aligned} \tag{17}$$

Solving this equation, two pairs of complex conjugate roots λ_1, λ_2, λ_1^*, λ_2^* (* denotes the complex conjugation) can be obtained. The structure-TMDI system is stable if both pairs of eigenvalues have negative real parts, i.e. the following condition applies

$$\max(real(\lambda_1,\lambda_2,\lambda_1^*,\lambda_2^*)) < 0 \tag{18}$$

The calculation of σ_{η_i} and $\sigma_{\eta_{TMDI}}$ then becomes a problem of constrained minimization. It can be achieved as follows: Firstly, in the first iteration step, a value of $\sigma_{\eta0}$ is assumed, the aeroelastic damping ξ_{ae} is then calculated by Eq. (15). Secondly, the transfer functions of the structure-TMDI system and the corresponding value of the RMS of the deck response $\sigma_{\eta1}$ are calculated through Eqs. (9)–(13), under the constrained condition as defined in Eq. (18). The difference between $\sigma_{\eta0}$ and $\sigma_{\eta1}$ is adopted as the optimization objective of this constrained minimization:

$$Op = \min(|\sigma_{\eta1}-\sigma_{\eta0}|) \tag{19}$$

If the tolerance is not satisfied, the value of $\sigma_{\eta1}$ is assigned to $\sigma_{\eta0}$, and the above procedure is repeated.

In the present study, the parameters defining the TMDI performance are: μ, β, f_{TMDI} and ξ_{TMDI}. The optimization design of TMDI in this study contains two steps: the first step is to choose the proper mass ratios (μ and β), and the second step is to find the optimal values of f_{TMDI} and ξ_{TMDI} based on the selected μ and β by minimizing the performance index J in Eq. (14).

To obtain the optimum values of f_{TMDI} and ξ_{TMDI}, the genetic algorithm (GA) is adopted in the present study. The genetic algorithm is a heuristic search method that is able to find the optimum solution of a problem with a termination criterion that either the maximum number of generation is reached or a satisfactory objective function is produced. Fig. 3 illustrates the diagram for the TMDI optimization. For a given set of the TMDI parameters (i.e. the individual chromosome in the right panel of Fig. 3), the RMS of the deck is calculated based on the above discussed constrained minimization algorithm. Then, the fitness for this

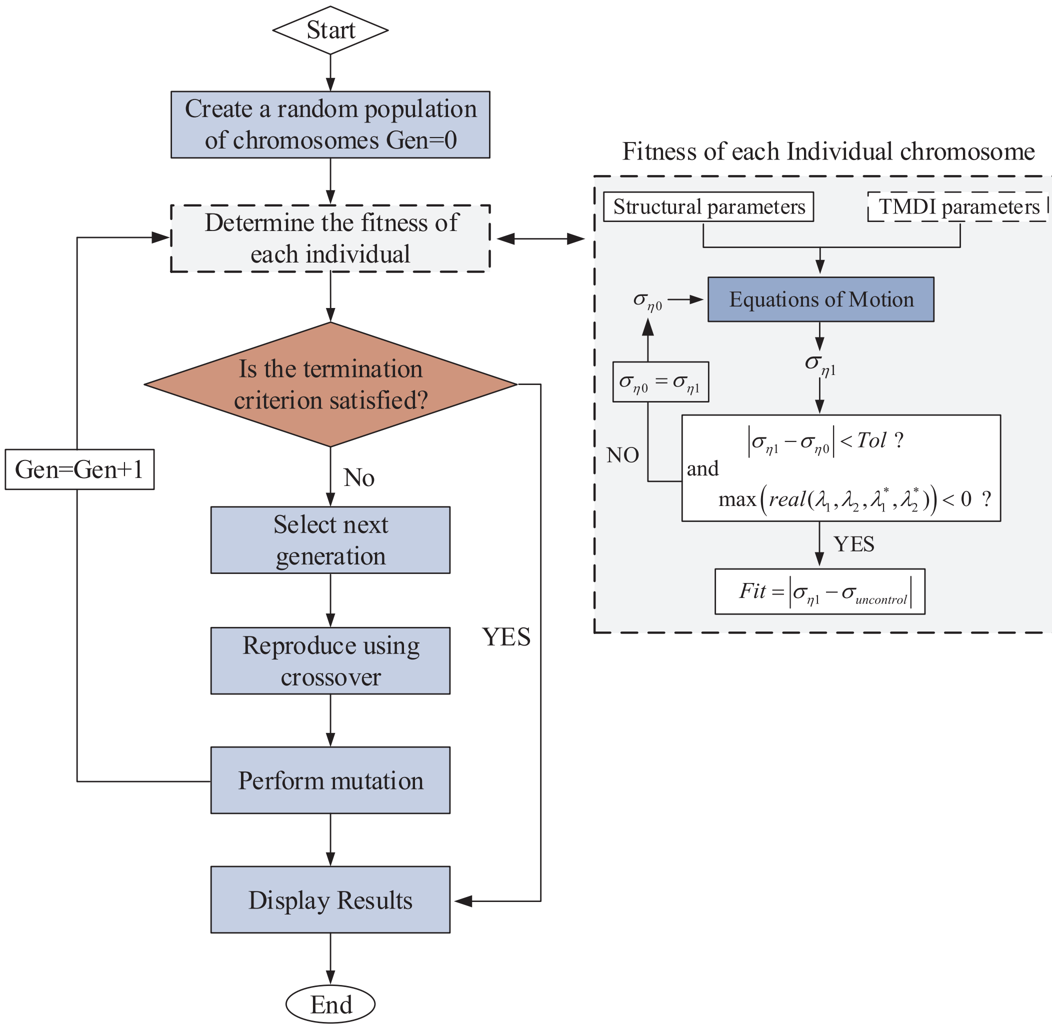


Fig. 3. Optimization diagram for the TMDI parameters.

individual chromosome can be obtained: $Fit = |\sigma_{\eta 1} - \sigma_{\eta i\_uncontrol}|$, where $\sigma_{\eta i\_uncontrol}$ denotes the RMS of the deck without TMDI. The optimum TMDI design parameters then correspond to the individual chromosome where the fitness reaches the minimum value. The constrained minimization algorithm and the genetic algorithm are conducted by using the "Optimization Toolbox" in the commercial software package MATLAB (version R2012a).

4. Numerical example

4.1. Bridge model

In this section, the effectiveness and performance of the TMDI system for bridge deck VIV control is investigated through a case study. The case to be investigated is a suspension bridge adopted from [40]. This bridge is adopted simply because the structural and aerodynamic parameters of this bridge are well defined in [40].

The main span of this bridge (between towers) is $L = 595$ m. The main deck of this bridge is a single cell steel box with a height of $D = 2.5$ m and width of 13.6 m. According to [40], this bridge experienced VIV at a vertical eigen-mode with the shape of four half-waves along the span between its two towers. The vibration amplitude is about 250 mm and the period is around 2.5 s. The mode shape function for this VIV mode is therefore defined as $\phi_i(x) = \sin(4\pi x/L)$ [40]. The modal frequency and modal damping ratio for this mode are $f_{ni} = 0.392$Hz and $\xi_{ni} = 0.24\%$, respectively [40]. The equivalent modal mass of the deck per unit length is $\bar{m} = 7460\text{kg}/m$.

The aerodynamic parameters for this bridge deck were determined through wind tunnel experiment in [40]. The nonlinear aeroelastic damping parameter a_L in Eq. (15) is assumed to be fixed for different wind velocities, and a value of 0.233 is adopted. While the linear aeroelastic damping parameter K_a varies with the wind velocity, which can be determined by:

$$\frac{K_a}{K_{a\max}} = \frac{0.9}{(U/U_{cr} - 0.25)^2}\cdot\exp\left[\frac{-1}{(U/U_{cr} + 0.02)^{24}}\right] - 0.18 \tag{20}$$

in which $U_{cr} = Df_{ni}/S_t$ is the critical wind velocity of VIV; $S_t = 0.16$ is the Strouhal number for the considered deck shape; $K_{a\max} = 2.43$ is the maximum value of K_a, which corresponds to the most unfavorable wind velocity of $U = 1.06U_{cr}$, at which the VIV response of the main deck reaches the maximum value. The load spectrum bandwidth parameter B for the vortex-shedding effect in Eq. (12) is chosen as 0.2, and the reduced RMS of the lift coefficient $\sigma_{CL}\cdot(\lambda/B)^{0.5}$ is taken as 3.92 [40]. The key structural properties of this bridge and the values of the aerodynamic parameters in Eqs. (12) and (15) are summarized in Table 1.

4.2. Optimized TMDI parameters

Similar to the conventional structure-TMD system, for the structure-TMDI system as shown in Fig. 2, the parameters to be optimized are the frequency (f_{TMDI}) and damping ratio (ξ_{TMDI}) of the system by following the optimization method as introduced in Section 3. As discussed in

Table 1
Structural and aerodynamic parameters for the case study.

L	D	f_{ni}	ξ_{ni}	$\bar{m}$	$\phi_i(x)$
595 m	2.5 m	0.392 Hz	0.24%	7460 kg/m	$\sin(4\pi x/L)$
$K_{a\max}$	a_L	$\frac{\sigma_{CL}^2\lambda}{B}$	B	S_t	
2.43	0.233	15.3664	0.2	0.16	

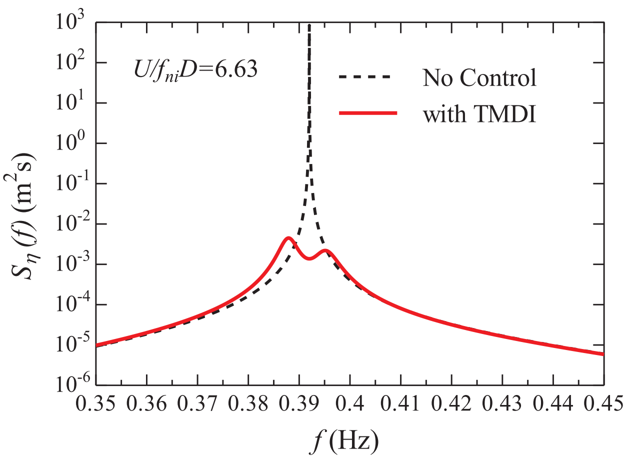


Fig. 4. Response spectra of the main deck before and after the installation of TMDI at the most unfavorable wind velocity.

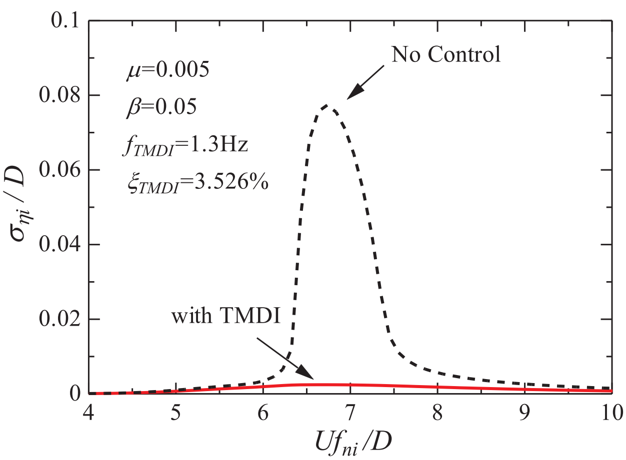


Fig. 5. RMS of the dimensionless main deck responses.

Section 3, the mass ratios μ and β as defined in Eq. (6) should be pre–determined. In this section, $\mu = 0.005$ and $\beta = 0.05$ are assumed. It should be noted that the selection of β is motivated by the fact that most current inerter devices can amplify the physical mass of the system by 60–200 times [22]. For example, when an inerter device with the capability of amplifying the physical mass by 100 times is selected, the physical mass of the inerter device is only 10% of the TMD device. As will be demonstrated in the following sections, the additional small mass can obviously enhance the control performance compared to the conventional TMD system.

Through the optimization procedure as discussed in Section 3 and the assumed mass ratios, the optimized values for f_{TMDI} and ξ_{TMDI} can be calculated as $f_{TMDI} = 1.3$ Hz and $\xi_{TMDI} = 3.526\%$. It is noticeable that the optimal frequency of the TMDI system is far from the structural natural frequency $f_{ni} = 0.392$Hz. This is different from the traditional TMD, in which the optimal frequency of the TMD system is around the structural natural frequency to make the TMD resonate with the primary structure. This result is in consistent with many previous studies where the optimal design frequency of TMDI increases with the inertance ratio β (e.g. [28,30]). This property actually makes the TMDI system more applicable compared to the TMD system when it is installed in the box cell of long-span bridges. This is because, as mentioned in the introduction, for a hanging-type TMD/TMDI, the static deformation of the spring due to gravity is $x_0 = g/\omega_{TMDI}^2$. For the selected example bridge, the static deformation of the spring is therefore 1.64 m and 0.148 m respectively if TMD and TMDI systems are installed. As mentioned in Section 4.1, the height of the bridge deck is only 2.5 m. If TMD system is installed, there will be almost no space for the TMD to deform considering the size of the device.

4.3. Effectiveness of the TMDI for VIV control

In this section, the effectiveness of using TMDI for bridge deck VIV control is investigated. Fig. 4 shows the response spectra of the main deck before and after the installation of the TMDI at the most unfavorable wind velocity within the lock-in region, at which the reduced velocity is $Uf_{ni}/D = 6.63$. As shown, without control device, the response spectra of the main deck show a one-peak characteristic, and this peak corresponds to the natural frequency of the primary structure. This is due to the fact that within the lock-in region the vortex-shedding frequency (the external forcing frequency) betrays the Strouhal law and is captured by the structural natural frequency [39]. After the TMDI is applied, the dominating component in the response spectra is significantly suppressed as shown by the solid line in Fig. 4. The response spectra of the main deck show a two-peak characteristic, and these peaks correspond to the frequencies of the structure and the TMDI system respectively-a two-degree-of-freedom (2-DOF) system as shown in Fig. 2.

Fig. 5 shows the dimensionless bridge deck response, i.e. the RMS of the deck responses with respect to the deck height σ_{η_i}/D, at different reduced wind velocities. It can be seen that at the most unfavorable wind velocity, the dimensionless response of the bridge deck reduces significantly from 0.085 to around 0.0025 when TMDI is installed, with a reduction ratio of 97%. The results also show that when the bridge structure is subjected to the other wind speeds, the TMDI system is also effective to reduce its vibration. These results clearly demonstrate the effectiveness of applying TMDI for bridge deck VIV control.

To further demonstrate the effectiveness of the proposed method, time domain analysis is also conducted. As discussed above, the aeroelastic damping ξ_{ae} in Eq. (15) is a function of σ_{η_i}, it is therefore cannot be directly used in the time domain calculations. On the other hand, ξ_{ae} can also be expressed as a function of A_{η_i} (the amplitude of deck response η_i) as follows [45]:

$$\xi_{ae} = K_a\frac{\rho D^2}{\bar{m}}\left[1-\left(\frac{\sigma_{\eta_i}}{Da_L}\right)^2\right] = K_a\frac{\rho D^2}{\bar{m}}\left[1-\left(\frac{A_{\eta_i}}{\sqrt{2}Da_L}\right)^2\right] \tag{21}$$

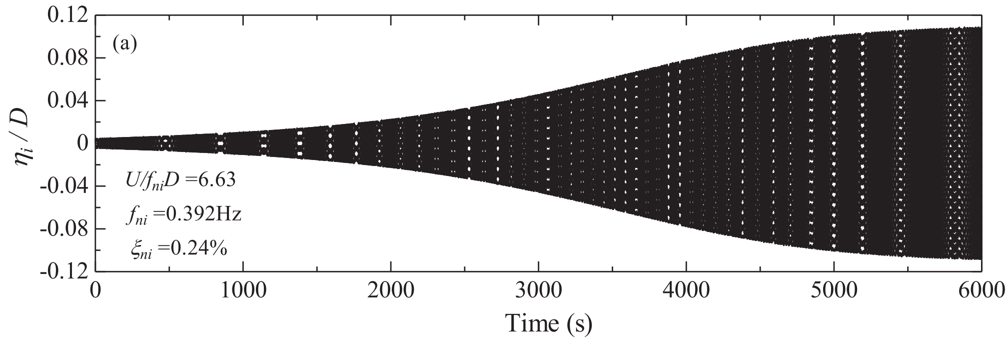
Then the harmonic balance technique can be adopted to transform the above expression into a function of the time-varying displacement of the main deck [45]:

$$\xi_{ae} = K_a\frac{\rho D^2}{\bar{m}}\left[1-4\left(\frac{\eta_i}{\sqrt{2}Da_L}\right)^2\right] = K_a\frac{\rho D^2}{\bar{m}}\left[1-2\left(\frac{\eta_i}{Da_L}\right)^2\right] \tag{22}$$

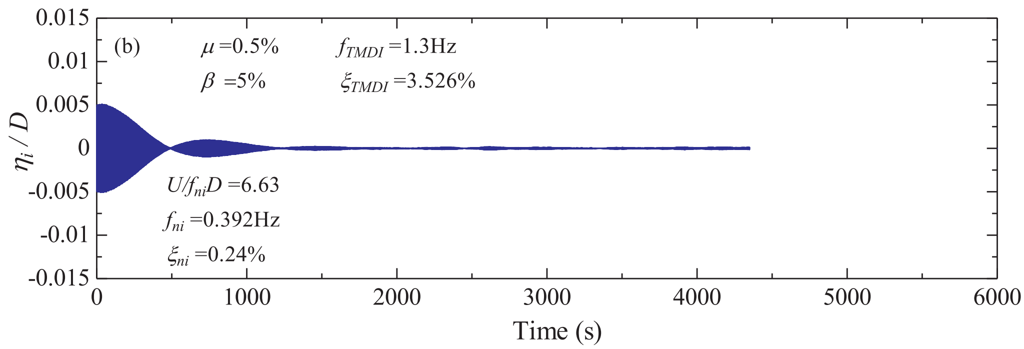
Comparing the above equation with Eq. (7), the following relations can be obtained:

$$\begin{cases} K_a = \frac{UY_1}{4\omega_{ni}D} \\ a_L = \frac{\sqrt{2}}{\sqrt{\varepsilon}}\sqrt{\frac{\int_0^L\phi_i^2(x)dx}{\int_0^L\phi_i^4(x)dx}} \end{cases} \tag{23}$$

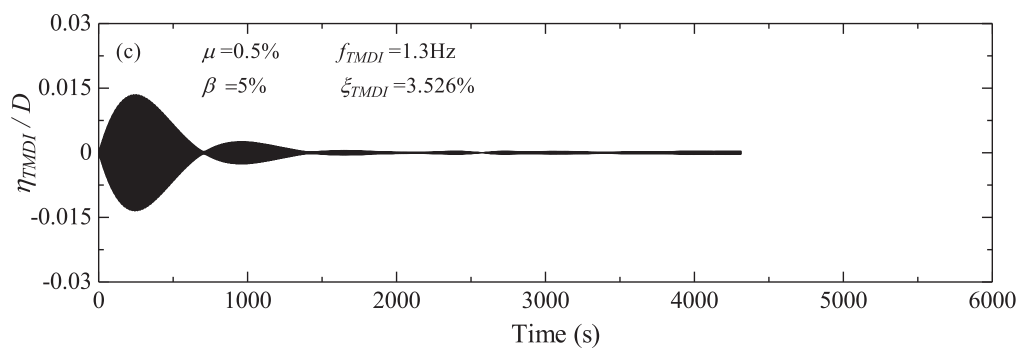
The displacements of the main deck and the TMDI then can be calculated by solving the following equations:



(a) main deck response without TMDI



(b) main deck response with TMDI



(c) relative displacement of the TMDI

Fig. 6. Time histories of the main deck with and without control.

$$\begin{Bmatrix}[\dot{\eta}]\\ [\ddot{\eta}]\end{Bmatrix} = \begin{Bmatrix}[\dot{\eta}]\\ [M]^{-1}[F]-[M]^{-1}[C][\dot{\eta}]-[M]^{-1}[K][\eta]\end{Bmatrix} \tag{24}$$

where

$$[M] = \begin{bmatrix} M_i + \mu M_i & \mu M_i \\ \mu M_i & \mu M_i + \beta M_i \end{bmatrix};$$

$$[C] = \begin{bmatrix} 2M_i\omega_{ni}(\xi_{ni} - \xi_{ae}) & \\ & 2\mu M_i \omega_{TMDI}\xi_{TMDI} \end{bmatrix};$$

$$[K] = \begin{bmatrix} M_i\omega_{ni}^2 - K_{ae} & \\ & \mu M_i \omega_{TMDI}^2 \end{bmatrix};$$

$$[\eta] = \begin{bmatrix} \eta_i \\ \eta_{TMDI} \end{bmatrix};$$

$$\text{and } [F] = \begin{bmatrix} F_i \\ 0 \end{bmatrix} \tag{25}$$

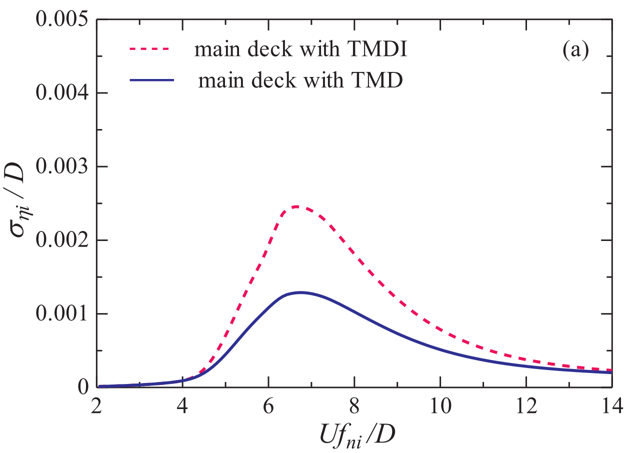
In Eq. (25), the time-varying vortex-shedding force F_i can be obtained by using the wave superimposition technique based on the PSD of the vortex shedding ($S_F(f)$ in Eq. (12)) [46]:

$$F_i(t) = \sum_{l=1}^{N} \sqrt{2\Delta f} H_v(f_l) cos(2\pi f_l t + \theta_l) \tag{26}$$

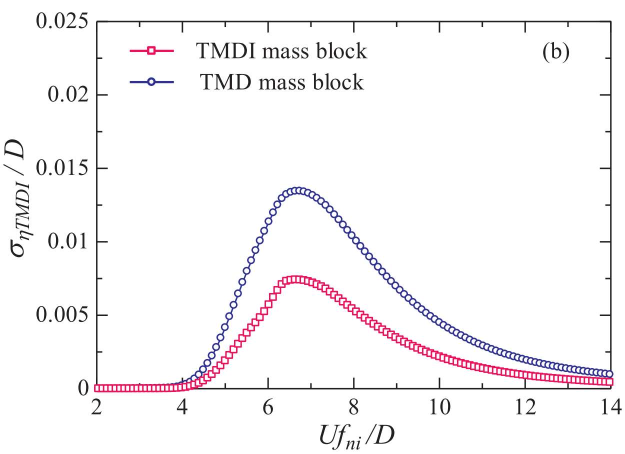
where $\Delta f = f_u/N$ is the frequency segment; N is the total number of frequency segments; f_u is the cut-off frequency; $f_l = l\Delta f$ is the lth frequency component; θ_l is a random phase angle that uniformly distributed within the range of $[0, 2\pi]$; and $H_v(f)$ is a lower triangular matrix obtained through the Cholesky's decomposition of $S_F(f)$.

In the calculations, the frequency interval is selected as $\Delta f = 0.01$Hz and the cut-off frequency is assumed as $f_u = 20$Hz. All the structural and aerodynamic parameters are the same with those in the frequency domain analysis.

Fig. 6 shows the displacements of the main deck and TMDI at the most unfavorable wind velocity (i.e. $Uf_{ni}/D = 6.63$). It can be seen from Fig. 6(a), for the uncontrolled case, the displacement of the bridge deck gradually increases from a small value to a large constant value at the steady-vibration state, i.e. the limit cycle oscillation, a typical vibration phenomenon induced by VIV, is formed. When TMDI is applied to the bridge deck, in the initial stage, the vibration of the deck leads to the resonance of the TMDI (large vibration of TMDI is therefore observed as

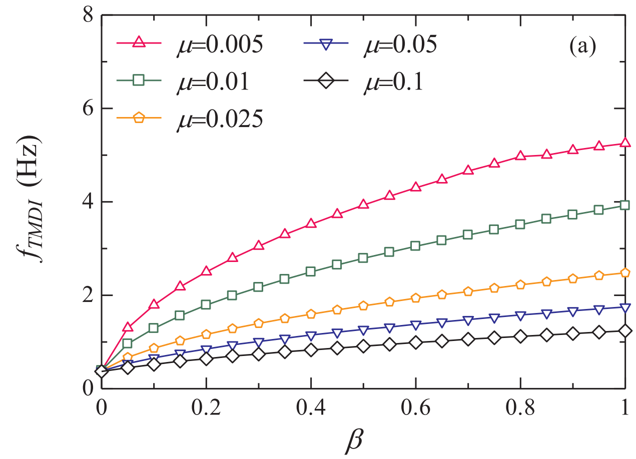


(a) normalized main deck response

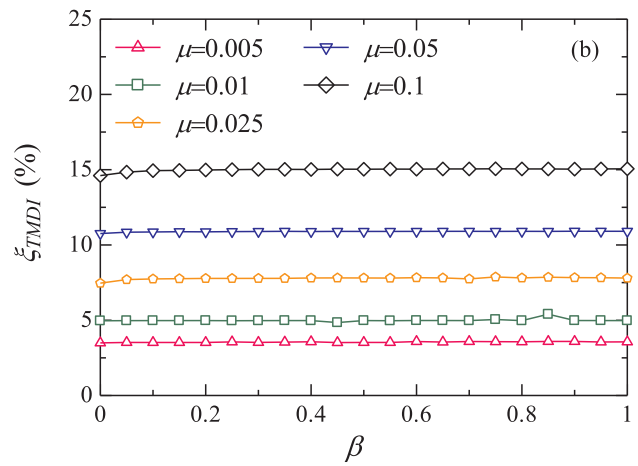


(b) normalized TMD/TMDI response

Fig. 7. Comparison between the TMDI and TMD systems.



(a) optimal frequency



(b) optimal damping ratio

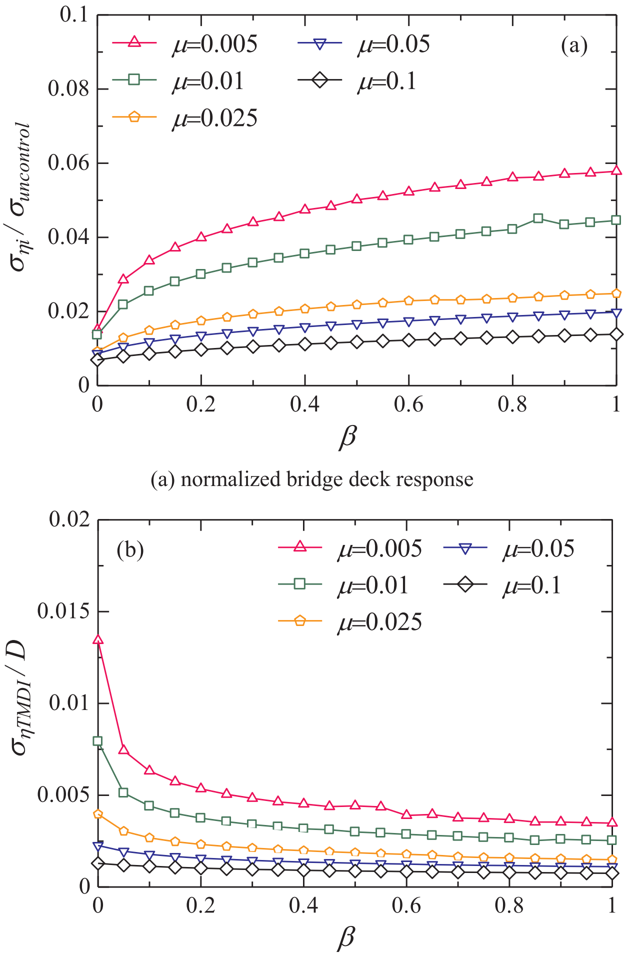
Fig. 8. Influence of μ and β on the optimal values of the TMDI design parameters.

shown in Fig. 6(c)), and the energy is transferred from the main deck to the TMDI and then dissipated by the damper. Thus, the amplitude of the main deck gradually decreases, as shown in Fig. 6(b). It should be noted that since the aeroelastic damping effect of VIV is motion-dependent, the energy transferred from the flow field to the structure-TMDI system decreases with the decrease of the deck amplitude, the vibrations of the main deck and TMDI therefore gradually vanish to a very small value after a period of time. Similar observations were obtained in many previous studies. For example in [14], dynamic control devices were applied to control the VIV of a bridge structure in Brazil, and the same phenomenon was obtained.

4.4. Performance comparison of TMDI and TMD systems

The performance of the TMDI system is compared with the conventional TMD system in this section. As mentioned in Section 2, the TMDI system degrades to the TMD system by setting $\beta = 0$. It is well known that for the TMD system, the frequency (f_{TMD}) and damping ratio (ξ_{TMD}) should be optimized in order to obtain the best performance of the system. For fair comparison, μ is assumed to be the same as that in the TMDI system (i.e. $\mu = 0.005$), and the other structural and aerodynamic parameters are also identical with those in the TMDI system. By applying the optimization technique as discussed in Section 3, $f_{TMD} = 0.39$ Hz and $\xi_{TMD} = 3.523\%$ are obtained. It is obvious that the optimal frequency of the TMD system is around the structural natural frequency so that the TMD resonates with the primary structure to dissipate energy.

Fig. 7(a) compares the dimensionless RMSs of the bridge deck responses and Fig. 7(b) shows the vibrations of the mass block in the TMDI and TMD systems. It can be seen that the maximum non-dimensional response of the main deck is around 0.0015 when TMD is installed, and the corresponding value becomes slightly larger (around 0.0025) if TMDI is applied. In other words, TMD system is more effective for VIV control compared to the proposed TMDI system. This is actually expected, since for the arrangement as shown in Fig. 2, the inclusion of the inerter prohibits the free oscillation of the mass block, therefore less energy will be dissipated by the TMDI system. Similar observation was reported in some previous studies. For example, Hu and Chen [47] compared the control performances of different inerter-based dampers. It was found that when the inerter was placed in parallel with the spring and damper of the conventional TMD (i.e. the same configuration of TMDI in this study), it provided no improvement to the control performance of conventional TMD. This effect is also clearly shown in Fig. 7(b), in which, the maximum dimensionless responses of the TMD and TMDI systems are 0.014 and 0.0075, respectively. The vibration of the TMD system is obviously larger than the TMDI system. On the other hand, it is worth reiterating again that the application of TMDI system in the box girder is more practical compared to the TMD system as discussed in Section 4.2. In other words, the proposed design makes the application more practical though the control effectiveness is slightly reduced. It should be noted that this trend is different from the ground-connection case (e.g. as adopted in [28,30]), in which one end of the inerter device is connected to the mass block while the other end is connected to the supporting ground. In this case, the relative



(a) normalized bridge deck response

(b) dimensionless TMDI vibration

Fig. 9. Influence of μ and β on the responses of the structure-TMDI system.

acceleration between the two terminals of the inerter is identical with that of the mass block since the supporting ground can be viewed as fixed. Thus, the resisting force generated by this device has an obvious effect of "amplification" on the mass block, and the TMDI system is more effective than the conventional TMD system with the same mass ratio.

5. Further discussions

5.1. Parametric study

In section 4, the effectiveness of using TMDI for bridge deck VIV control is demonstrated for particular mass ratios μ and β. In this section, the influences of μ and β on the control effectiveness are further investigated through a parametric study. For each μ and β, the optimal values of the TMDI system are obtained by using the procedure as presented in Section 3.

Fig. 8(a) and 8(b) show the optimal TMDI frequency and damping ratio respectively. As shown, with the increase of the mass ratio μ, the optimal frequency decreases (Fig. 8(a)), while the optimal damping ratio increases (Fig. 8(b)). This result is in consistent with the conventional TMD system, and the same trend was also reported in many other literatures (e.g. [28,30]) when the inerter device was applied to control earthquake induced vibrations of engineering structures. Fig. 8(a) also shows that for a particular μ, the optimal frequency increases with the increment of the inertance ratio β. In this sense, the involving of the inerter device tends to amplify the optimal frequency as compared to the conventional TMD system, and this effect is more obvious for the relatively small β as the slope of the curve decreases with the increase of β, as shown in Fig. 8(a). This property is actually beneficial for the design as discussed above, since the larger frequency can result in a smaller static deformation of the spring in the TMD/TMDI system. For the damping ratio, the influence of β is, however, not evident. As shown in Fig. 8(b), for a particular μ, the optimal damping ratio is almost a constant though β varies obviously. This indicates that no external damping is required when the inerter device is incorporating into the conventional TMD system, which makes the additional cost not significant.

Fig. 9(a) shows the influences of μ and β on the normalized VIV responses of the bridge deck. In particular, the RMS of the main deck displacement obtained from the TMDI controlled system is normalized by the value from the uncontrolled system. It can be seen that the application of TMDI can significantly reduce the VIV response of the main deck since all the values are less than 6% for the considered μ and β. In other words, more than 94% of the VIV response is suppressed when the TMDI is applied. The results also show that the control effectiveness is more evident with larger μ. This result is in consistent with the conventional TMD again. For a particular μ, the control effectiveness slightly decreases with the increase of β. This is because the involvement of the inerter device constrains the free oscillation of the TMD device as explained above.

Fig. 9(b) shows the non-dimensional responses of the TMDI mass block with respect to the deck height. Similar to the conventional TMD system again, the vibration of the mass block decreases with the increase of the mass ratio μ. For a particular μ, larger β results in smaller TMDI mass block vibrations, and this effect is more evident when β is relatively small. This will further reduce the required space and in turn makes the TMDI system more feasible compared to the TMD system.

5.2. Sensitivity analysis

The TMD/TMDI system will be most effective when the optimal frequency and damping are used, and these parameters are obtained based on the structural and aerodynamic properties of the bridge. However, some uncertainties inevitably exist in real engineering practices. The robustness of the system is therefore important. The robustness of the TMD/TMDI system is investigated through a sensitivity analysis in this section.

Fig. 10 shows the variation of the control performance of the main deck ($\sigma_{\eta_i}/\sigma_{uncontrol}$) against different design parameters (ξ_{TMDI} and f_{TMDI}) for various values of μ and β. The Z-axis in these figures is revised in order to better illustrate the shape of the control performance surface. As shown, the 3D surfaces show long and narrow shapes, which means that the control performance can be obviously influenced by perturbing the TMDI frequency, while changing the damping ratio of the TMDI system, the control performance changes little. In other words, the control performance of the structure-TMDI system is more sensitive to the TMDI frequency while less sensitive to the TMDI damping.

Fig. 10 also indicates that when the mass ratio μ becomes larger, the widths of the convex surfaces versus TMDI frequency become wider, which indicates that the TMDI system with a larger mass ratio μ is more robust than that with a smaller value. This result is in consistent with the conventional TMD system again. On the other hand, when the mass ratio μ is the same, the band widths of the convex surfaces with different β are almost the same and they are similar to that of the convention TMD system (with$\beta = 0$), which means the sensitivity of the TMDI system is mainly governed by the TMD system, and the influence of inertance (β) on the system robustness is not obvious.

6. Concluding remarks

This paper proposes using TMDI to control the VIV of the main deck of long-span bridges. The equation of motion of the structure-TMDI

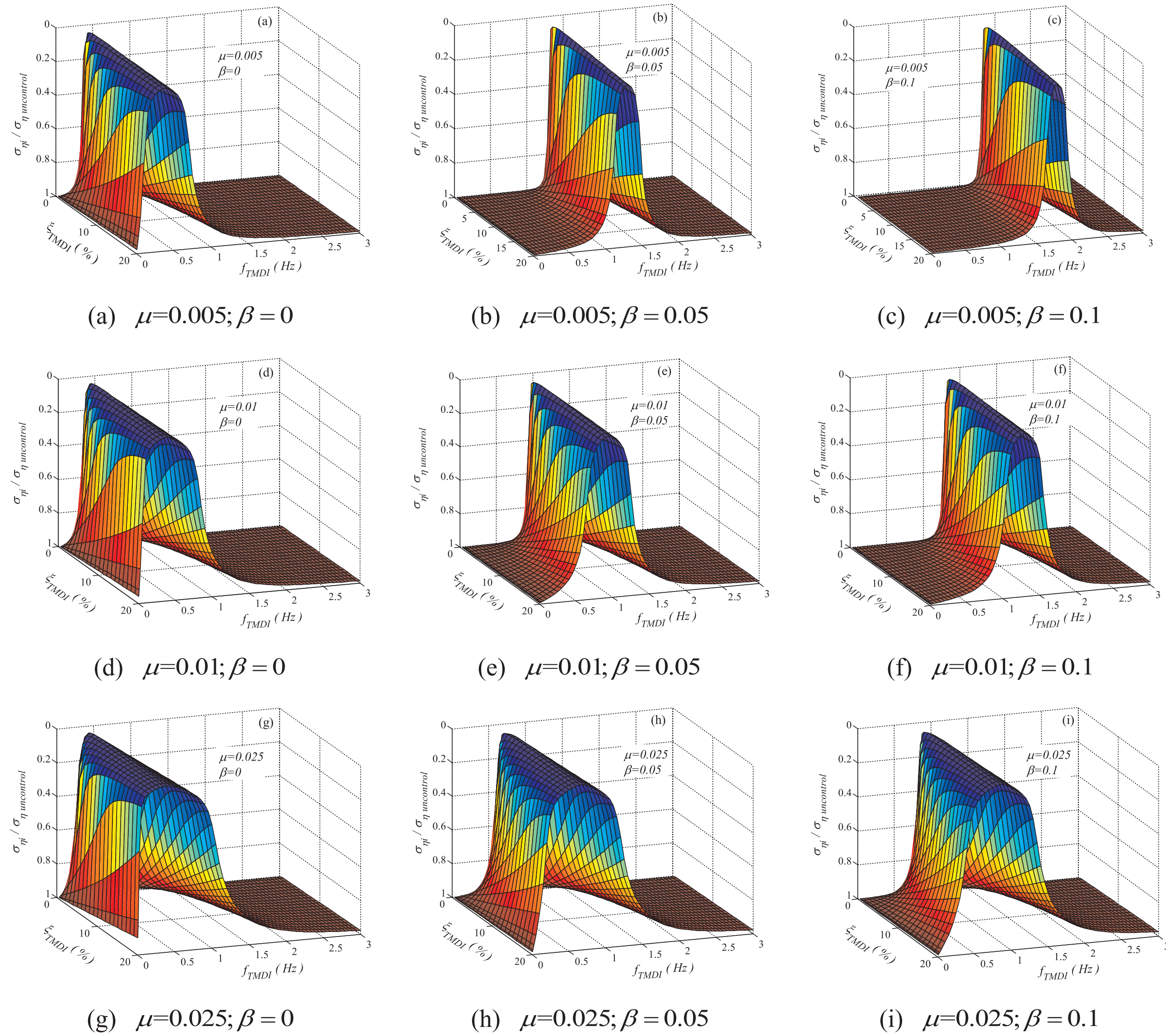


Fig. 10. Influence of μ and β on the intensity contour of the main deck displacement.

system subjected to VIV is derived by taking the nonlinear aeroelastic effect into consideration. The optimal parameters of the TMDI system are calculated and the effectiveness of the TMDI system is evaluated. Following conclusions are obtained based on the analytical results:

(1) The TMDI system can significantly reduce the VIV of the main deck. More than 94% of the vibration of the main deck of the example bridge is suppressed when TMDI is applied.
(2) Compared to the TMD system, the effectiveness of the TMDI system is slightly reduced. However, applying TMDI system for VIV control is more practical compared to the TMD system since the TMDI system can significantly reduce the static deformation and oscillation of the mass block in the system.
(3) The control effectiveness of the TMDI system is more evident with larger μ, while for a particular μ, the control effectiveness decreases slightly with the increase of β.
(4) The control performance of the TMDI system is more sensitive to the TMDI frequency while less sensitive to the TMDI damping.
(5) The robustness of the TMDI system is mainly governed by the TMD, the influence of inertance on the system robustness is not obvious.

It should be noted that the proposed method can be applied not only to the new constructions but also to the existing bridge structures with conventional TMD systems. All these properties make the proposed TMDI system an attractive alternative for the VIV control of long-span bridges.

Acknowledgements

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References

[1] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay Crossing Bridge. J Struct Eng 2002;128:1012–25.
[2] Li H, Laima S, Ou J, Zhao X, Zhou W, Yu Y, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. Eng Struct 2011;33:1894–907.
[3] Frandsen JB. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. J Wind Eng Ind Aerodyn 2001;89:95–129.
[4] Larsen A, Esdahl S, Andersen JE, Vejrum T. Storebælt suspension bridge – vortex shedding excitation and mitigation by guide vanes. J Wind Eng Ind Aerodyn 2000;88:283–96.

[5] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. J Wind Eng Ind Aerodyn 2014;124:54–67.
[6] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. Struct Eng Int 2013;23:248–68.
[7] Murakami T, Takeda K, Takao M, Yui R. Investigation on aerodynamic and structural countermeasures for cable-stayed bridge with 2-edge I-shaped girder. J Wind Eng Ind Aerodyn 2002;90:2143–51.
[8] Zhou R, Yang Y, Ge Y, Mendis P, Mohotti D. Practical countermeasures for the aerodynamic performance of long-span cable-stayed bridges with open decks. Wind Struct An Int J 2015;21:223–39.
[9] Dale JC, Irwin PA, King JPC. Recent developments concerning vortex-induced oscillations of edge girder bridges. Struct Congr 2006:1–10.
[10] Larsen A, Savage M, Lafrenière A, Hui MCH, Larsen SV. Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. J Wind Eng Ind Aerodyn 2008;96:934–44.
[11] Ge Y, Yang Y, Cao F. VIV sectional model testing and field measurement of Xihoumen Suspension Bridge with twin box girder. In: The 13th international conference on wind engineering. Amsterdam, Netherlands; 2011.
[12] Fujino Y, Kimura K, Tanaka H. Wind resistant design of bridges in Japan: developments and practices. Tokyo: Springer; 2012.
[13] Sarwar MW, Ishihara T. Numerical study on suppression of vortex-induced vibrations of box girder bridge section by aerodynamic countermeasures. J Wind Eng Ind Aerodyn 2010;98:701–11.
[14] Battista RC, Pfeil MS. Reduction of vortex-induced oscillations of Rio-Niterói bridge by dynamic control devices. J Wind Eng Ind Aerodyn 2000;84:273–88.
[15] Wardlaw RL. The improvement of aerodynamic performance. In: Proceedings of the first international symposium on aerodynamics of large bridges. Copenhagen, Denmark; 1992.
[16] Weber F, Distl J, Maślanka M. Semi-active TMD concept for Volgograd Bridge. Topics in dynamics of civil structures. New York: Springer; 2013.
[17] Seo JW, Kim HK, Jin P, Kim KT, Kim GN. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. J Wind Eng Ind Aerodyn 2013;116:7–20.
[18] Domaneschi M, Martinelli L, Po E. Control of wind buffeting vibrations in a suspension bridge by TMD: hybridization and robustness issues. Comput Struct 2015;155:3–17.
[19] Martinelli L, Domaneschi M. Effect of structural control on wind fatigue mitigation in suspension bridges. Int J Struct Eng 2017;8:289–307.
[20] Li C, Zhou D. Evaluation of multiple active lever-type tuned mass dampers for structures underground acceleration. Eng Struct 2004;26:303–17.
[21] Smith MC. Synthesis of mechanical networks: the inerter. IEEE Trans Autom Control 2002;47:1648–62.
[22] Papageorgiou C, Smith MC. Laboratory experimental testing of inerters. Proceedings of the 44th IEEE conference on decision and control, and the European Control Conference. Seville, Spain. 2005. p. 3351–6.
[23] Li C, Liang M, Wang Y, Dong Y. Vibration suppression using two terminal flywheel Part II: application to vehicle passive suspension. J Vib Control 2012;18:1353–65.
[24] Wang FC, Chen CW, Liao MK, Hong MF. Performance analyses of building suspension control with inerters. In: Proceedings of the 46th IEEE conference on decision and control. New Orleans, LA, USA2007. p. 3786-91.
[25] Wang FC, Hong MF, Chen CW. Building suspensions with inerters. Proc IMechE, J Mech Eng Sci 2010;224:1605–16.
[26] Ikago K, Saito K, Inoue N. Seismic control of single-degree-of-freedom structure using tuned viscous mass damper. Earthquake Eng Struct Dyn 2012;41:453–74.
[27] Lazar IF, Neild SA, Wagg DJ. Using an inerter-based device for structural vibration suppression. Earthquake Eng Struct Dyn 2014;43:1129–47.
[28] Marian L, Giaralis A. Optimal design of a novel tuned mass-damper–inerter (TMDI) passive vibration control configuration for stochastically support-excited structural systems. Probab Eng Mech 2014;38:156–64.
[29] Wen Y, Chen Z, Hua X. Design and evaluation of tuned inerter-based dampers for the seismic control of MDOF structures. J Struct Eng 2016;143:04016207.
[30] Pietrosanti D, Angelis MD, Basili M. Optimal design and performance evaluation of systems with Tuned Mass Damper Inerter (TMDI). Earthquake Eng Struct Dyn 2017;46:1367–88.
[31] Lazar IF, Neild SA, Wagg DJ. Vibration suppression of cables using tuned inerter dampers. Eng Struct 2016;122:62–71.
[32] Sun L, Hong D, Chen L. Cables interconnected with tuned inerter damper for vibration mitigation. Eng Struct 2017;151:57–67.
[33] Shi X, Zhu S. Dynamic characteristics of stay cables with inerter dampers. J Sound Vib 2018;423:287–305.
[34] Giaralis A, Petrini F. Wind-induced vibration mitigation in tall buildings using the tuned mass-damper-inerter. J Struct Eng 2017;143:04017127.
[35] Ma R, Bi K, Hao H. Mitigation of heave response of semi-submersible platform (SSP) using tuned heave plate inerter (THPI). Eng Struct 2018;177:357–73.
[36] Smith MC, Wang F-C. Performance benefits in passive vehicle suspensions employing inerters. Veh Syst Dyn 2004;42:235–57.
[37] Wang F-C, Liao M-K, Liao B-H, Su W-J, Chan H-A. The performance improvements of train suspension systems with mechanical networks employing inerters. Veh Syst Dyn 2009;47:805–30.
[38] Wang F-C, Hong M-F, Lin T-C. Designing and testing a hydraulic inerter. Proc Inst Mech Eng, Part C: J Mech Eng Sci 2011;225:66–72.
[39] Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. J Eng Mech 1990;116:1392–411.
[40] Strommen E, Hjorth-Hansen E. On the use of tuned mass dampers to suppress vortex shedding induced vibrations. Wind Struct An Int J 2001;4:19–30.
[41] Basu RI, Vickery BJ. Across-wind vibrations of structure of circular cross-section. Part II. Development of a mathematical model for full-scale application. J Wind Eng Ind Aerodyn 1983;12:75–97.
[42] Vickery BJ, Basu RI. Across-wind vibrations of structures of circular cross-section. Part I. Development of a mathematical model for two-dimensional conditions. J Wind Eng Ind Aerodyn 1983;12:49–73.
[43] Yozo F, Masato A. Design formulas for tuned mass dampers based on A perturbation technique. Earthquake Eng Struct Dyn 1993;22:833–54.
[44] Gattulli V, Di Fabio F, Luongo A. Simple and double Hopf bifurcations in aeroelastic oscillators with tuned mass dampers. J Franklin Inst 2001;338:187–201.
[45] Chen X. Estimation of stochastic crosswind response of wind-excited tall buildings with nonlinear aerodynamic damping. Eng Struct 2013;56:766–78.
[46] Shinozuka M, Jan CM. Digital simulation of random processes and its applications. J Sound Vib 1972;25:111–28.
[47] Hu Y, Chen MZQ. Performance evaluation for inerter-based dynamic vibration absorbers. Int J Mech Sci 2015;99:297–307.